



AI & Machine Learning

AI grew up and got a job: Lessons from 2025 on agents and trust

December 18, 2025



Will Grannis

VP and CTO, Google Cloud

Google Cloud experts review how 2025
transformed AI from simple chatbots into

In our Ask OCTO column, experts from Google Cloud's Office of the CTO answer your questions about the business and IT challenges facing you and your organization now. Think of this series as Google Cloud's version of an advice column — except the relationships we're looking to improve are the ones in your tech stack.

This month, we're looking back at 2025 with Will Grannis and his team at the Office of the CTO. As the year comes to a close, we asked the team to reflect on the most significant AI insights, lessons, and developments of the year.

Understanding agents



AI grew up and got a job

Antonio Gulli

2025 was the year we stopped chatting with AI and started treating it like an actual employee. The difference between an LLM and an agent is huge.

An LLM is a brain in a jar that knows facts. An agent is that same brain with hands and a plan. It uses

Take a party planning agent. Where a traditional chatbot would give you a guacamole recipe, an agent checks your calendar to pick a date, emails friends to see who's free, orders avocados through a grocery store API, and creates a Spotify playlist based on what your friends like. It does the work instead of talking about it.

To keep agents from going off the rails, developers can use Agentic Design Patterns: Guardrails block risky actions, critics review outputs for errors before you see them, and routers direct different parts of complex tasks to specialized models.



Moving beyond atomic tools

Yingchao Huang

As we move from deterministic software to probabilistic agentic workflows, we face a critical reliability gap. Traditional databases rely on ACID properties to prevent corruption. Current agent frameworks often model multi-step actions as continuous flows without a transaction coordinator (TC), creating non-atomic failure modes. When an agent crashes mid-operation — paying a vendor before updating a record — it risks irreversible side effects and data corruption.

implement patterns like “agent undo stacks” and TCs that encapsulate complex logic into atomic, reversible units. Using mechanisms like idempotent tools and checkpointing, failures trigger safe rollbacks rather than leaving inconsistent states. This shifts the reliability burden from the probabilistic LLM to deterministic system design, where it belongs.



Agents that learn on the job

Michael Zimmermann

In 2025, agents arrived center stage with everyone expecting meaningful ROI. But agents designed by engineers often lack the tribal knowledge gained through experience in finance, legal, HR, and sales.

On-the-job agent learning is being heavily researched: how agents evolve after launch, learn, and adapt. Agents' quality and trust can grow meaningfully during deployment and don't need to score 100% on all metrics day one. Agents can shadow experts, learn and evolve.

Agents can perform to a certain level initially, equipped with building blocks that let them rapidly evolve and sometimes surpass humans. The critical piece is creating the learning loop where agents take signals from the environment and humans in



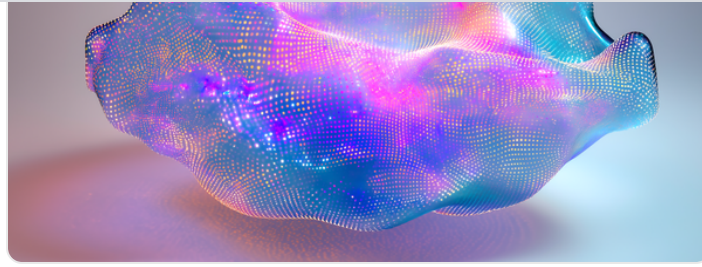
The trust integration challenge

Troy Trimble

This year AI reached a point where it can handle greater volume and complexity while agents can increasingly automate routine work. This means we can focus on higher-value intellectual tasks.

Integrating agents into existing workflows is both crucial and challenging because humans lack complete trust in AI. This trust deficit requires robust processes allowing gradual integration of AI functionalities as we build confidence. Second, agent identity and its integration with existing human and service account-based identity systems is still developing. These concepts need refinement before widespread adoption.

After successful integrations build trust, we'll transition to AI-first systems where human involvement remains a core design principle, given equal priority alongside AI user journeys.



The ROI of AI

Leading companies are moving beyond debating AI's possibilities and are now using AI agents to drive measurable business value and achieve significant ROI by automating complex workflows. Discover how in our new report.

[Read it now.](#)

Deploying at scale



The shift to edge inference

Pablo Rodriguez Rodriguez

The conversation around AI fundamentally shifted from capability to trust in 2025. For our largest global customers, trust is defined by sovereignty (control over data, infrastructure, and models) and compliance (adherence to regulations like the EU AI Act).

workloads surpassed training workloads this year.

As agents and AI content generation exploded, the challenge became secure, cost-effective service at scale for sovereign customers where most activity happens at the edge.

One breakthrough was enabling Gemini to run securely on-premise in Google Distributed Cloud environments. This required extending confidential computing to the edge, ensuring that sensitive data and model weights are encrypted while being actively used. Secure enclaves using hardware-level memory encryption made this possible.



Simulation before production

Hann Wang

In 2025, AI moved from experimental pilots to broad adoption. As the industry has evolved, deploying agents has become less a software problem and more a governance challenge.

In complex workflows with multiple agents, it's difficult to isolate which agent drove success or caused failure. Static benchmarks like Q&A tests are obsolete for business use cases. Real business is dynamic, adversarial, and negotiated.

complex scenarios. This allows us to stress-test strategic thinking and adaptability in a safe sandbox, ensuring agents are battle-tested before they interact with live customers.

We also applied game theory to evaluation pipelines, establishing a mathematical framework for credit attribution. This gives enterprises KPI-level visibility to audit AI performance, reward the right behaviors, and correct risks.

Building responsibly



Every GenAI project is an eval project

Ben McCormack

After working on GenAI projects globally, one pattern became clear: every GenAI project rapidly becomes an evaluation project.

Evaluation differs from classical software engineering where we define exact outputs for given inputs. GenAI is nuanced: an LLM might give different answers each time, and customers often deploy chatbots as gateways to their companies,

This means teams need to become experts in evaluation for all GenAI projects. Successful teams invest in building reusable evaluation scaffolding that lets them measure quality consistently as models and requirements evolve.



Evaluation as real-time self-correction

Carina Claassen

Evaluation evolved from a passive metric to an active architectural component in 2025. Integrating evaluation directly into agentic pipelines as a closed-loop system is a powerful strategy for improving quality.

An autorater (an LLM acting as judge) assesses each agent output in real-time. When it detects an error, it provides actionable feedback the agent uses to retry or correct itself, steering toward better outcomes without human intervention.

This self-correction mechanism solves the compounding error problem. When an agent makes a mistake in step two, traditional evaluation only catches it after step ten fails. Real-time autoraters catch and fix errors at the source, before they cascade.

them for higher overall quality. This closed-loop approach levels up evaluation from a static quality measurement to a powerful tool for dynamic quality improvement.



Business leaders must learn the KPIs for AI

Chuck Sugnet

The metrics determining AI success deserve the same attention as revenue or EBITDA, but most executives don't understand them yet.

Trying to manage AI projects without AI KPIs ends as poorly as running a business without accounting. Consider fraud detection: an algorithm predicting every transaction as "not fraudulent" would be 99% accurate if only 1% of transactions are fraudulent, but would catch zero fraud. This introduces precision (correctness of fraud predictions) and recall (percentage of actual fraud caught).

Business leaders need to adopt these measurement frameworks now. Three investments matter most:

- Build measurement literacy across teams
- Treat evaluation datasets as strategic assets
- Embed continuous measurement into operations so teams can iterate quickly



Bias toward action

Jen Bennett

When technology moves this fast, it's tempting to wait for perfection. Don't. Dig in and use the opportunity to learn.

Some projects were too early and didn't work as well as we hoped. Those lessons became fuel for the next wave of innovation. We learned to share findings generously, early, and often. We retest when new models come out and stay comfortable throwing away code and pivoting quickly.

We spend significant time thinking about evaluation: How will we know when something performs well? What metrics matter most? Like test-driven development, the best place to start is with how you plan to evaluate "good."

Practical craft



Art directing your GenAI

The most crucial lesson from 2020: the quality of AI output is a direct reflection of the quality of instruction. We've moved past treating AI like a magic black box. The most effective leaders are directing it.

To master generative AI, we must understand how it processes instructions. For text generation, LLMs are prediction engines. They take your prompt and predict the most probable next word. Vague instructions force the model into a wide search space, leading to generic responses.

For image generation, your prompt converts into a mathematical embedding that guides the process. Vague words like "good" give the model no clear visual direction.

In both cases, precision is your most powerful lever. Act as the Art Director by defining scope, style, and structure before the model starts generating. This eliminates ambiguity that leads to generic results. Specificity is cheaper and more effective than regenerating outputs until something works.



The four ingredients for success

Diane Chaleff

transformative outcomes:

- A **use case** that matters to the business. Make sure the topic you're applying AI to actually matters.
- **Data.** Without realistic data, you can't know if an experiment will actually work. Gathering and refining the right data is often one of the longest steps.
- **Metrics.** How do you know if each experiment run is getting better or worse? You need clear, climbable, explainable metrics. Many AI use cases live in gray areas without yes/no outcomes. Spend time deciding how you'll measure.
- **Appropriate error risk.** Even the best-designed AI isn't flawless. What steps can your solution take to limit risk? Can you put boundaries on acceptable output? Are end users knowledgeable enough to notice errors? This question requires self-judgment but matters before starting.



AI accelerating science

Jeff Sternberg

For me, 2025 was the year of AI advancing science. I came to this through weather forecasting, helping to launch WeatherNext in Google Cloud. As the year

Modern science relies on large datasets and scaled data analysis. General purpose models like Gemini handle both structured and unstructured scientific sources, making them perfect for agentic AI systems like AI Co-Scientist.

In Co-Scientist, agents search and understand scientific literature and datasets. Other agents generate novel research ideas based on this research combined with the scientist's prompt. Still other agents review generated ideas and score them for novelty, feasibility, and impact. Similar to LMArena, Co-Scientist runs a tournament with AI agent judges, simulating peer review. This produces ELO scores that let the most promising research ideas bubble to the top.

This process accomplishes research that would otherwise take scientists days or weeks. One colleague told me, referring to AI agents for science: "this really works." If you know any scientists, that's high praise.



Vibe coding and understanding

Mark Schadler

functions and gained the capability to understand and interact with entire codebases. This unlocked new powers of AI-assisted development, including "vibe understanding" as an effective way to learn and explore even large codebases.

While fundamental development principles still apply (it's easier to write new code than read and maintain old code), AI-assisted development is decreasing the distance between these two activities.

This represents a shift in how developers work. Instead of spending hours tracing through unfamiliar code, developers can now ask questions and get coherent explanations of system design, data flows, and integration points. The ability to have a conversation with your codebase fundamentally changes the development experience.

Conclusion

Three things defined 2025: agents got jobs, evaluation became architecture, and trust became the bottleneck.

The technical progress was real. Task-optimized tools replaced fragile API chains. Learning loops let agents improve after launch. Sovereign cloud infrastructure extended to the edge with confidential computing making it viable. Cultural

teams need to develop comfort deploying imperfect systems that improve over time.

Successful AI deployment requires building the infrastructure to deploy systems that learn, the evaluation frameworks to measure improvement, and the trust mechanisms to integrate AI into workflows gradually. In 2025, we saw the start of these shifts at enterprise scale.

Posted in [AI & Machine Learning](#)

Related articles



AI & Machine Learning

The "infinite capacity" myth: How AI is breaking the old cloud rules

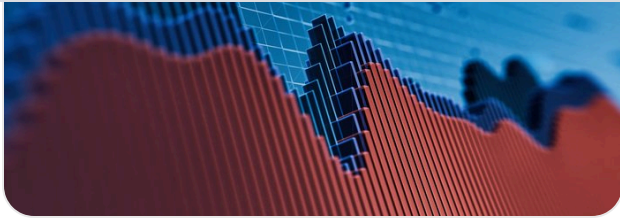
By Jamie de Guerre • 3-minute read



AI & Machine Learning

Your agents are leaving the building

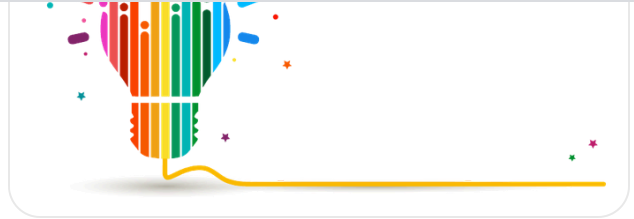
By Will Grannis • 6-minute read



AI & Machine Learning

The KPIs that actually matter for production AI agents

By Benazir Fateh • 9-minute read



AI & Machine Learning

Beyond the pilot: Five hard-won lessons from Google Cloud's AI transformation

By Amy Liu • 10-minute read

Follow us



Google Cloud

Google Cloud Products

Privacy

Terms



Help

English